

# Using Machine Learning to Identify No-Show Telemedicine Encounters in a New York City Hospital

Wanting CUI<sup>a,1</sup> and Joseph FINKELSTEIN<sup>a</sup>

<sup>a</sup> *Icahn School of Medicine at Mount Sinai, New York, NY, USA*

**Abstract.** No-show visits are a serious problem for healthcare centers. It costs a major hospital over 15 million dollars annually. The goal of this paper was to build machine learning models to identify potential no-show telemedicine visits and to identify significant factors that affect no-show visits. 257,293 telemedicine sessions and 152,164 unique patients were identified in Mount Sinai Health System between March 2020 and December 2020. 5,124 (2%) of these sessions were no-show encounters. Extreme Gradient Boosting (XGB) with under-sampling was the best machine learning model to identify no-show visits using telemedicine service. The accuracy was 0.74, with an AUC score of 0.68. Patients with previous no-show encounters, non-White or non-Asian patients, and patients living in Bronx and Manhattan were all important factors for no-show encounters. Furthermore, providers' specialties in psychiatry and nutrition, and social workers were more susceptible to higher patient no-show rates.

**Keywords.** No-show visits, supervised machine learning, telemedicine

## 1. Introduction

No-show patients are defined as patients making an appointment with the healthcare centers, but failing to attend their appointments without previous notice. No-show visits is a common and important problem for hospitals not only in the United States but several countries around the world [1]. It could cost a major hospital over 15 million dollars annually [2]. Although various methods had been proposed to reduce the no-show rate, such as reminder systems and imposing penalization, these efforts were limited. The average no-show rate for a healthcare center was 3% to 18% based on various factors [2]. Thus, building predictive models to identify potential no-show patients is important. Several previous studies had successfully constructed predictive models and identified important factors that are associated with no-show visits [1, 3-5]. However, all of these studies only considered in-person visits. Telemedicine visits could be different, as it eliminates the transportation constraints, but imposes higher requirements for technology. Thus, the goal of this study was to build machine learning models to identify potential no-show patients in telemedicine visits and to identify significant factors that affect no-show visits.

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<sup>1</sup> Wanting Cui, Icahn School of Medicine at Mount Sinai, 1770 Madison Ave, 2<sup>nd</sup> Fl, New York, NY, USA, 10035; E-mail: wanting.cui@mssm.edu.

## 2. Method

The dataset was extracted from the electronic health record (EHR) at Mount Sinai Health. We identified all telemedicine visits between March 2020 and December 2020. We defined telemedicine visits as video visits, telehealth, telephone visits, telemedicine visits and non-face to face visits. We separated the dataset into two groups: patients that didn't show up for the visit and patients presented at the visit. Since there was limited information to extract from patients who didn't show up for their appointments, we identified 10 factors that could be obtained prior to their arrivals. The 10 variables were visit type, age, sex, race, New York City Boroughs, health providers' primary specialty, providers' type, day of the week, number of previous telemedicine visits and number of previous no-show encounters.

Patients' sex was limited to male and female, and race was categorized into White, Black, Asian and others. We mapped patients' zip code into boroughs. We further identified the top 5 most frequent providers' type and specialties from each subset. The top most frequent provider types were physician, social worker, resident, nurse practitioner and psychologist. And the top most frequent provider primary specialties were internal medicine, adult psychiatry, children psychiatry, neurology, gastroenterology, family medicine, nutrition, urology and pediatric care. We also mapped patients' appointment date into day of the week. Since each patient could have multiple encounters, we treated each encounter independently.

We aimed to build a machine learning model to identify potential no-show encounters based on their appointment information. In previous studies, the most common method to evaluate no-show prediction was logistic regression, followed by decision trees [1]. In our previous paper, we explored the effectiveness of logistic regression and tree based models on imbalanced medical data prediction, and tree based model with sampling achieved the best result [6]. Since no-show encounters only constitute 2% of the dataset, we explored various sampling techniques and sampling ratio to balance the imbalanced dataset. We compared 3 non-parametric machine learning models, support vector machine (SVM), random forest (RF) and extreme gradient boosting (XGB). We performed parameter tunings and performed 3-fold cross validation to find the best parameters. The predictive dataset is randomly divided into 70% training and 30% testing. Since the data is imbalanced, we used Area under the ROC curve (AUC) as our evaluation metric. We applied the best-tuned model for each algorithm to the testing set and calculated accuracy, AUC accordingly.

## 3. Results

There were 257,293 telemedicine sessions between March 2020 and December 2020. 5,124 of telemedicine session were no-show encounters, whereas 252,169 encounters patients showed up. There were 152,164 unique patients in the dataset and 4,150 patients had at least one no-show encounter during this time period.

In machine learning models, the 10 predictors were all information that could be obtained when patient make an appointment with their healthcare provider or through previous encounter histories. The target variable was whether a patient presented to the telemedicine session. There were 2 classes, no-show and present. We explored 3 sampling techniques, random under sampling, random up sampling and synthetic minority oversampling technique (SMOTE). Both random under sampling and random

up sampling performed well. However, with under sampling, the number of records generated through this method were around 12,000 records, which was significantly less than the 400,000 records generated through up sampling. Thus, we chose under sampling which decreased the models training times significantly.

In SVM, we tuned the regularization parameter (C) and kernel coefficient (gamma). The final test set accuracy was 0.75, with AUC of 0.64. In RF, we tuned the number of trees (n\_estimators), number of features to consider at each split (max\_features), maximum number of levels in trees (max\_depth) and minimum number of samples (min\_samples\_leaf) required to split a node. The best model generated an accuracy of 0.81, with AUC of 0.66. In XGB, we tuned maximum depth of a tree (max\_depth), step size shrinkage (eta), minimum loss reduction required (gamma), and minimum sum of instance weight needed in a child (min\_child\_weight). The best model generated an accuracy of 0.74, with AUC of 0.68. Although RF model provided the best accuracy, its AUC score was lower than that of XGB model. Thus, we chose XGB model as our final model. 954 out of 1520 (63%) of no-show encounters were correctly identified from the imbalanced dataset, and 56,168 out of 75668 (74%) of encounters were correctly identified as patients who would show up.

Table 1. Results of machine learning models.

| Model | Sampling | Parameter                                                                                           | Accuracy | AUC  |
|-------|----------|-----------------------------------------------------------------------------------------------------|----------|------|
| SVM   | RUS      | C=10, gamma=0.01                                                                                    | 0.75     | 0.64 |
| RF    | RUS      | max_depth=10, max_features='sqrt',<br>min_samples_leaf=2, min_samples_split=5,<br>n_estimators=1200 | 0.81     | 0.66 |
| XGB   | RUS      | eta=0.1, gamma=0.1, max_depth=5,<br>min_child_weight=5                                              | 0.74     | 0.68 |

We further investigated the feature importance of XGB model. We identified the top 5 factors: patients' previous no-show encounters, race and boroughs, providers' type and specialty. From patients' information (Table 2), we found that 20% no-show encounters happened to patients' with previous no-show records. Patients who were self-identified as Black or other race were significantly more likely to not show up for their telemedicine visits, compared to White and Asian patients. In addition, patients who lived in Manhattan and Bronx had a higher level of no-show rate, and patients who lived outside New York City region had a low level of no-show rate.

Table 2. Top features affecting patients' no-show rate based on patients' information

|                         | No Show Encounters |         | Present Encounters |         |
|-------------------------|--------------------|---------|--------------------|---------|
|                         | count              | percent | count              | percent |
| <b>Previous no show</b> |                    |         |                    |         |
| 0 times                 | 4171               | 81.40%  | 245999             | 97.60%  |
| 1-2 times               | 605                | 11.80%  | 5142               | 2.00%   |
| 3 or more times         | 348                | 6.80%   | 1028               | 0.40%   |
| <b>Race</b>             |                    |         |                    |         |
| Asian                   | 269                | 5.20%   | 15126              | 6.00%   |
| Black                   | 1077               | 21.00%  | 31392              | 12.40%  |
| Others                  | 2253               | 44.00%  | 87517              | 34.70%  |
| White                   | 1525               | 29.80%  | 118134             | 46.80%  |
| <b>Borough</b>          |                    |         |                    |         |

|           |      |        |       |        |
|-----------|------|--------|-------|--------|
| Bronx     | 658  | 12.80% | 18916 | 7.50%  |
| Brooklyn  | 757  | 14.80% | 42537 | 16.90% |
| Manhattan | 2155 | 42.10% | 87279 | 34.60% |
| Others    | 923  | 18.00% | 75508 | 29.90% |
| Queens    | 631  | 12.30% | 27929 | 11.10% |

Furthermore, from providers' perspectives, nutritionists, psychologists and social workers faced significantly higher no-show rates compared to other providers. In contrast specialty care providers, especially those in cardiology, dermatology, and pediatric care had significantly less no-show rates.

4. Discussion

XGB was the best model, it had the highest AUC score (0.68) compared to SVM and RF models. In addition, XGB model could provide feature importance that allowed us to analyze factors that are associated with no-show encounters. Patients with previous no-show encounters, non-White or non-Asian patients and patients lived in Bronx and Manhattan were all important factors for no-show encounters. It's important to acknowledge that although telemedicine service didn't require patients to travel to hospital or clinics, patients' residential location was still a significant factor in the model. Furthermore, providers' specialty in psychiatry and nutrition, and social workers were more susceptible to higher patient no-show rate.

5. Conclusions

XGB with under sampling was the best machine learning model to identify no-show patients using telemedicine service. Furthermore, patients' previous no-show encounters, race and boroughs, providers' type and specialty were the 5 factors that were highly correlated to no-show encounters.

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